



GOVERNMENT POLYTECHNIC, AMRAVATI
(An Autonomous Institute Of Government of Maharashtra)

CURRICULUM DEVELOPMENT CELL

DIPLOMA PROGRAMME: COMPUTER ENGINEERING

COURSE CODE: CM5460

COURSE TITLE: COMPUTER GRAPHICS

TEACHING SCHEME:

LEVEL OF COURSE	PRERE-QUISITE	WEEKLY CONTACT HRS.			TOTAL CREDITS	TOTAL WEEKS	TOTAL CONTACT HOURS		
		L	T	P			L	T	P
V	FC3401	03	-	02	05	16	48	-	32

EXAMINATION SCHEME:

THEORY(Marks)					PRACTICAL(Marks)		TOTAL (Marks)
ESE PAPER HRS.	ESE		PA	TOTAL	ESE	PA	
3Hrs.	MAX.	70	30*	100	25#	25^	150
	MIN.	28	-	40	10	10	

@: Internal Assessment #: External Assessment- Practical base

(*) Under the Theory PA, Out Of 30 Marks, 20 Marks is the Average of Two Tests and 10 Marks are for Micro project.

(^) Under practical PA Continuous Assessment of Practical Work is to be done by Course Teacher as per CDC norms.

For the courses having only practical examination, PA has two parts (i) Continuous Assessment of Practical work - 60% and (ii) microproject-40%.

1.RATIONALE:

Almost every computer system is set up to allow the user to interact with the system through a graphical user interface, where information on the display screen is conveyed in both textual and graphical forms. Everyone should be aware of this rapidly expanding technology.

Computer graphics is a complex and diversified technology. The output product of Computer graphics is a pictorial image. Hence the computer has become a new tool for the artist and animator. Computer graphics is an extremely effective medium for communication between man and machine through pictures, charts and diagrams.

To use images, pictures in any systems one needs the understanding of generation, editing, processing and conversion of it into other types of images.

Computer graphics techniques can be used in many fields such as Engineering drawing, business graphs, architectural design and also for video games, which provides a new form of entertainment.

2.COURSE OUTCOMES:

At the end of the course , students will be able to:-

1. Identify the file structure of display graphics file formats.
2. Apply the algorithms to draw lines, circles & polygons.
3. Use transformation techniques to scale, rotate & translate the object.
4. Draw the line clipping & polygon clipping using algorithm.
5. Develop the logic for drawing the natural objects using different algorithms for curved lines.
6. Illustrate the fundamentals of interactive graphics.

3. DETAILED CONTENTS: THEORY

Unit	Unit Outcomes (UOs) (In cognitive domain)	Topic and Sub-topics	CO No.	Hour	Marks
Unit 1 Basics of CG	1a. Differentiate attributes of the given mode. 1b. Compare features of the given scan display. 1c. Describe application of the given display device. 1d. Write a program to draw the given type of primitives using "C". 1e. Convert the given 2D coordinates to physical device coordinates.	1.1 Introduction to computer graphics, advantages of computer graphics, applications of computer graphics. 1.2 Display devices: Raster scan display, random scan display, Flat panel display, LED display, LCD display, plasma display, touch screen. 1.3 Graphics functions and standards. Types of graphics file formats: General explanation, advantages & disadvantages. 1.4 Latest trends in Computer Graphics. Virtual reality, Augmented reality. 1.5 Co-ordinate systems (2D, 3D): cartesian, Polar.	1	06	08
Unit 2 Line, Circle, and Polygon.	2a. Write a program to draw line using the given algorithm. 2b. Use the given algorithm to rasterize the given line. 2c. Apply the given algorithm to generate the circle. 2d. Draw the polygon using the given algorithm. 2e. Apply character	2.1 Basic concepts in line drawing, line drawing algorithms: DDA algorithm, Bresenham's algorithm. 2.2 Circle generating algorithms: DDA circle drawing algorithm, Bresenham's circle drawing algorithm,	2	10	14

	generation method to display the given character.	midpoint circle algorithm. 2.3 Polygons – Types of polygons, Inside Outside test, polygon filling: Seed fill algorithms: flood fill, boundary fill, scan-line algorithms. 2.4 Sample problems to illustrate above algorithms.			
Unit 3 Transformations	3a. Perform the given operation in 2D transformation. 3b. Perform the given operation in 3D transformation. 3c. Solve the given problem based on composite transformations. 3d. Apply the given type of projection on object.	3.1 2D transformation: Translation, Rotation, Scaling, Reflection, Shearing. 3.2 Matrix Representation and Homogeneous Co-ordinates: Translation, Rotation, Scaling, Reflection and Shearing. 3.3 Composite Transformation: Rotation about an arbitrary Point. 3.4 3D Transformations: Translation, rotation, Scaling. 5 Sample problems with sample coordinates to illustrate above algorithms.	3	10	14
Unit 4 Windowing & Clipping	4a. Apply window to viewport transformation on the given object. 4b. Write a program using the given line clipping algorithm. 4c. Apply the given line clipping algorithm to	4.1 Windowing & Clipping Concepts: Window to Viewport transformation. 4.2 Line clipping: Cohen-Sutherland Subdivision Line clipping	4	10	14

	clip the line. 4d. Write a program using the given polygon clipping algorithm.	algorithm, Midpoint subdivision line Clipping algorithm. 4.3 Polygon clipping: Sutherland – Hodgeman Polygon clipping algorithm. 4.4 Sample problems to illustrate above algorithms.			
Unit5 Curves and Fractals	5a. Describe the given curve generation methods. 5b. Draw curve using the given curve algorithms. 5c. State properties of the given curve. 5d. Generate arc using the given algorithm.	5.1 Curve generation: Arc generation using DDA algorithm, Interpolation. 5.2 Types of Curves: Bezier Curve, Hilbert's curve, Koch curve. 5.3 Fractals: Fractal lines, Fractal surfaces.	5	08	12
Unit 6 Interactive Graphics	6a. Apply the graphics standards 6b. Describe the graphical user interface. 6c. Describe the principles for good GUI design. 6d. Describe OpenGL.	6.1 Need for graphics standards, Advantages of Graphics standards, Hazards of Graphics standards. 6.3 Graphical user interface: Introduction, Example. 6.4 Features of GUI, Principles for good GUI design. 6.5 Open GL: What is Open GL, How OpenGL works.	6	04	08

4. LIST OF PRACTICALS:

S.N.	PRACTICAL OUTCOMES (PrOs)	CO NO.
1	Write a program to draw following graphics objects using built in “C” functions. Pixel Lines Circles Rectangle Ellipse	1
2	Implement following algorithms to draw a line. DDA algorithm. Bresenham’s algorithm.	2
3	Implement following algorithm’s to draw a circle (Any Two). DDA algorithm. Bresenham’s algorithm. Midpoint circle algorithm.	2
4	Write a program to fill polygon using following methods: i) Flood fill ii) Boundary fill	2
5	Write a program for two dimensional transformation (Any Two) Translation Scaling Rotation	3
6	Write a program for three dimensional transformation Translation Scaling.	3
7	Write a program to clip line using Cohen-Sutherland algorithm.	4
8	Write a program to clip line using Cohen Midpoint subdivision algorithm.	4
9	Write a program to clip polygon using Sutherland – Hodgeman algorithm.	4
10	Implement Arc generation using DDA algorithm.	5
11	Write a program to draw following type of curves (Any Two). Hilbert’s curve Koch curve Bezier curve	5
12	Write a program to develop Tic-Tac-Toe game using OpenGL.	6

Note

- i) The entire above listed practical’s need to be performed compulsorily, so that the students reaches the ‘Precision level of Dave’s Psychomotor Domain Taxonomy’.
- ii) The Process and Product related skills associated with each practical outcome shall be assessed on the basis of following performance indicators.

S. No.	Performance Indicators	Weightage in %
1.	Write program to draw graphics objects	20
2.	Use graphics software tool for programming to create, edit, compile the programs/applications	20
3.	Debug, test & execute the programs/applications	20
4.	Completed the exercise in stipulated time	20
5.	Able to answer oral questions	10
6.	Submit report in time	10
Total		100%

The above PROs also comprise of the following social skills/attitudes which are Affective Domain Outcomes (ADOs) that are best developed through the laboratory/field based experiences:

- Handle command prompt environment.
- Experiment with graphics environment.
- Plan, construct, compile, debug and test programs.
- Maintain tools and equipment.
- Follow ethical Practices.

The ADOs are not specific to any one PrO, but are embedded in many PrOs. Hence, the acquisition of the ADOs takes place gradually in the student when s/he undertakes a series of practical experiences over a period of time. Moreover, the level of achievement of the ADOs according to Krathwohl's 'Affective Domain Taxonomy' should gradually increase as planned below:

- 'Valuing Level' in 1st year
- 'Organizing Level' in 2nd year and
- 'Characterizing Level' in 3rd year

5. SUGGESTED STUDENT ACTIVITIES

Other than the classroom and laboratory learning, following are the suggested student-related co-curricular activities which can be undertaken to accelerate the attainment of the various outcomes in this course: Students should conduct following activities in group & prepare a report of about 5 pages for each activity, also collect/record physical evidences for their (student's) portfolio which will be useful for their placement interviews:

- Prepare journals based on practical performed in laboratory.
- Draw perspective and parallel projection for any object on view plane.
- Give seminar on relevant topic .
- Prepare power point presentation or animation for showing different types of graphics applications.
- Undertake a market survey of different graphics application and compare with the following points.
 - Available Applications.
 - Application Profile.

6. SUGGESTED INSTRUCTIONAL STRATEGIES

These are sample strategies, which the teacher can use to accelerate the attainment of the various outcomes in this course:

- Massive open online courses (MOOCs) may be used to teach various topics/sub topics.
- The teacher needs to ensure to create opportunities and provisions for co-curricular

activities.

- c. About 10-15% of the topics/sub-topics which is relatively simpler or descriptive in nature is to be given to the students for self-directed learning and assess the development of the COs through classroom presentations. Keep the record of the topics/sub-topics/ contents given to the students.
- d. Guide student(s) in undertaking micro-projects.
- e. Demonstrate students thoroughly before they start doing the practice.
- f. Continuously observe and monitor the performance of students in Lab.

7. SUGGESTED MICRO-PROJECTS.

Only one micro-project is planned to be undertaken by a student assigned to him/her in the beginning of the semester. S/he ought to submit it by the end of the semester to develop the industry oriented COs. Each micro-project should encompass two or more COs. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The micro project could be industry application based, internet-based, workshop-based, laboratory-based or field-based. Each student will have to maintain dated work diary consisting of individual contribution in the project work and give a seminar presentation of it before submission. The total duration of the micro-project should not be less than 16 (sixteen) student engagement hours during the course. In all semesters, the micro-project could be group-based to build up the skill and confidence in every student to become problem solver so that s/he contributes to the projects of the industry.

A suggestive list is given here. Similar micro-projects could be added by the concerned faculty:

- a) Program to Design Flying balloons.
- b) Program to display a moving car .
- c) Develop a miniature tic-tac-toe game.
- d) Design an analog clock.
- e) Design a rotating fan.
- f) Design a smiling face.
- g) Design a scientific calculator.
- h) Design a traffic signal system.
- i) Design a solar system.
- j) Design arrival and departure of train.
- k) Design a sinking ship.

8. MAJOR EQUIPMENTS/INSTRUMENTS REQUIRED

S.N.	Equipment Name with Broad Specification	Practical No.
1	Hardware: Personal computer (i3-i5 preferable), RAM minimum 2GB onwards.	For all practicals
2	Operating System: Windows XP / Windows 7 / LINUX onwards	
3	Software: Turbo C with with DOSBOX or Emulated C.	

9. SUGGESTED SPECIFICATION TABLE FOR QUESTION PAPER DESIGN

Unit No	Unit Title	Marks per units	Distribution of theory marks		
			R level	U level	A level
1.	Basics of CG	08	4	4	--
2.	Line, Circle & polygon	14	2	4	8
3.	Transformations	14	2	4	8
4.	Windowing & Clipping	14	2	8	4
5.	Curves & Fractals	12	4	4	4
6.	Interactive Graphics	08	4	4	--
	Total	70	18	28	24

R=Remember, U=Understand, A=Apply and above (Bloom's Revised taxonomy)

Note: This specification table provides general guidelines to assist student for their learning and to teachers to teach and assess students with respect to attainment of UOs. The actual distribution of marks at different taxonomy levels (of R, U and A) in the question paper may vary from above table.

10. LEARNING RESOURCES:

SrNo	TITLE	AUTHOR	PUBLICATION
1	Computer Graphics	Donald Hearn, Baker M. Pauline	Pearson Education ,New Delhi June 2012, ISBN:817758765X
2	Computer Graphics	Maurya Rajesh K.	Wiley-India 2011, Delhi ISBN:978-81-265-3100-4
3	Computer Graphics	Dr. Chopra Rajiv	S. Chand 2016, New Delhi. ISBN:978-93-856-7633-8.
4	Computer Graphics principles & practices	Foley James	Pearson Education ,New Delhi 2014, ISBN:978-0-321-39952-6

11. SOFTWARE/LEARNING WEBSITES.

- www.tutorialspoint.com
- www.nptelvideos.in
- www.dailyfreecode.com
- www.newtechnologysite.com
- www.w3schools.com
- www.codeacademy.com
- www.howstuffworks.com
- www.computerhope.com

12. COURSE CURRICULUM DEVELOPMENT COMMITTEE:

S.N.	NAME	DESIGNATION	INDUSTRY/INSTITUTE
1	Smt. S. R. Patil	Lecturer in Computer Engineering	Government Polytechnic Amravati
2	Miss. S. S. Chavan	Lecturer in Computer Engineering	Government Polytechnic Amravati
3	Smt. C. P. Ahir	Lecturer in Computer Engineering	Government Polytechnic Amravati

Government Polytechnic, Amravati Programme Board of Studies Computer Engineering has approved the above course curriculum on 14/05/2020 and is adopted for Computer Engineering Programme.

CHAIRMAN
PROGRAMME BOARD OF STUDIES,
COMPUTER ENGG DEPARTMENT
GOVERNMENT POLYTECHNIC, AMRAVATI

The General Board of Studies has approved the above course curriculum on 06/02/2021

The Governing Body has approved the above course curriculum on 13/08/2021.